Daystak Side Chair RD1



1:4 Scale Model Project

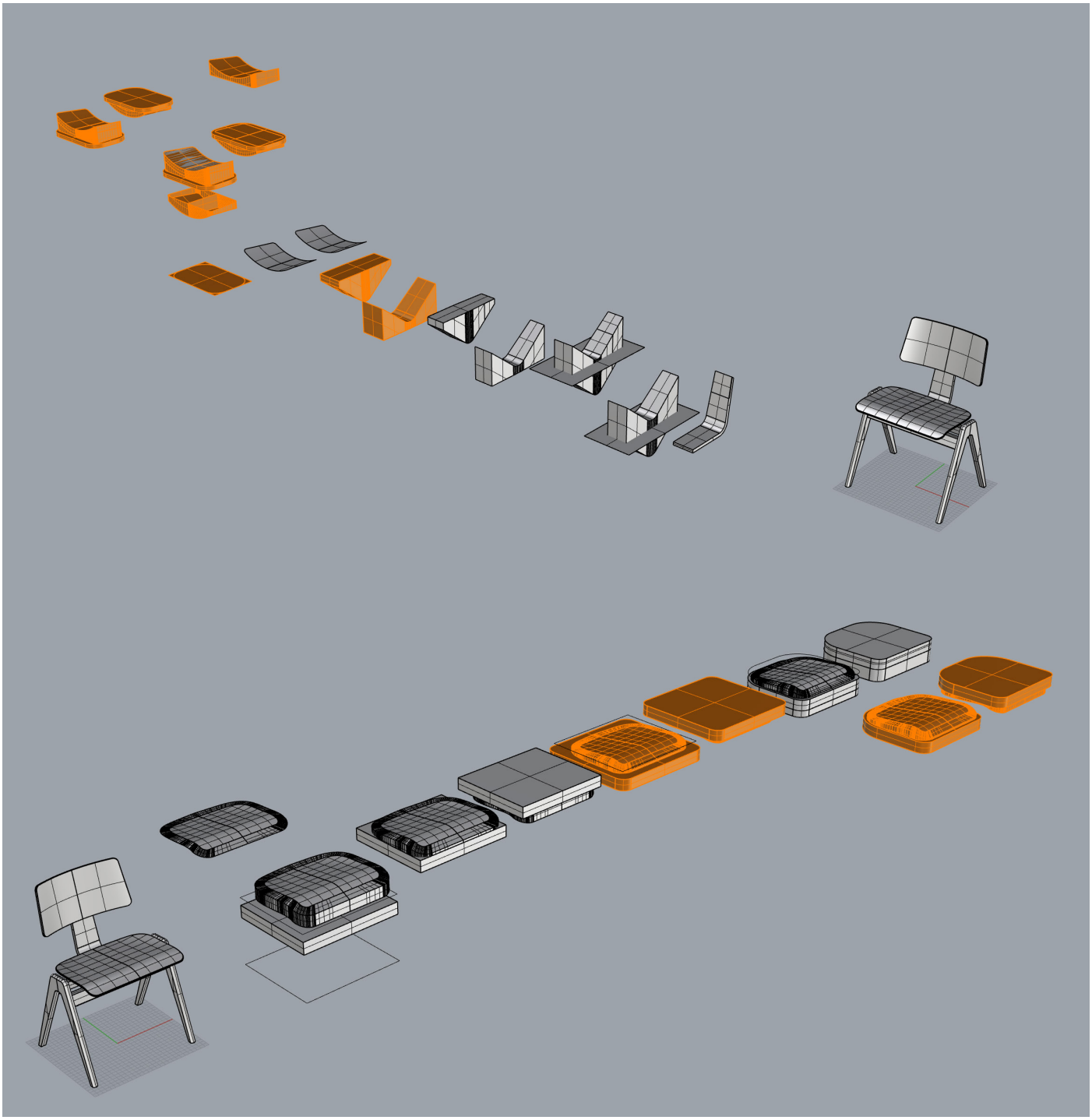
Making Scale Model



Creating Moulds

Step 1

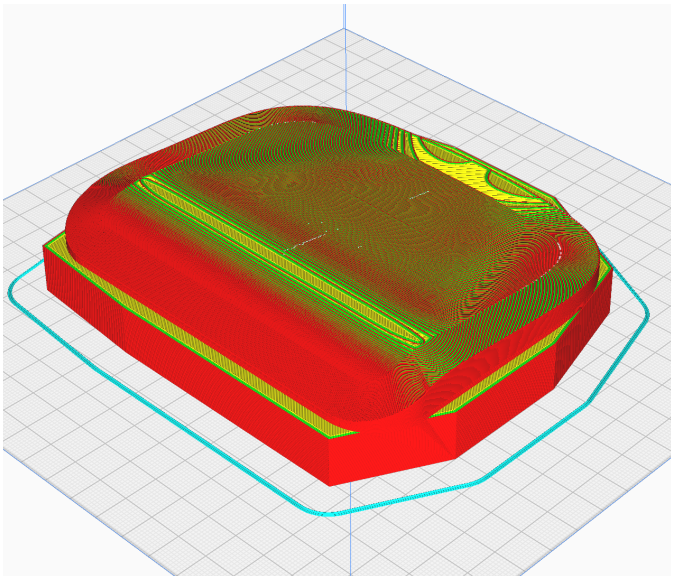
Creating moulds to press-form the backrest and the seat-plate



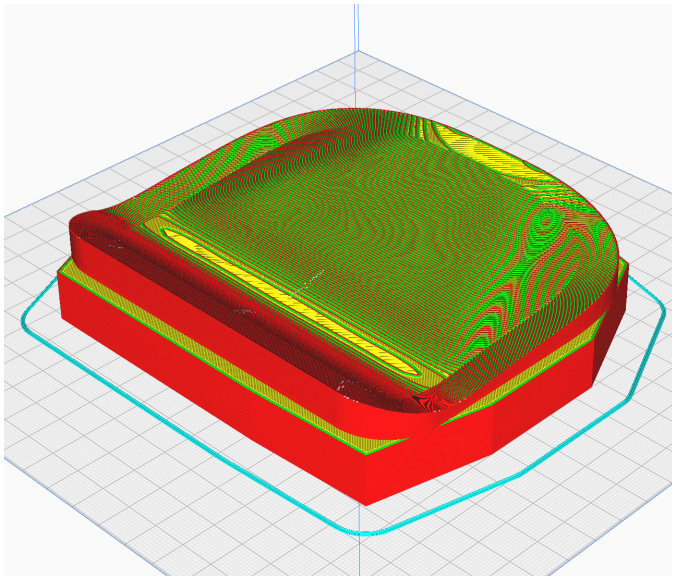
Process of making molds in rhino

Step 2

Preparing the 3d file for 3d printing



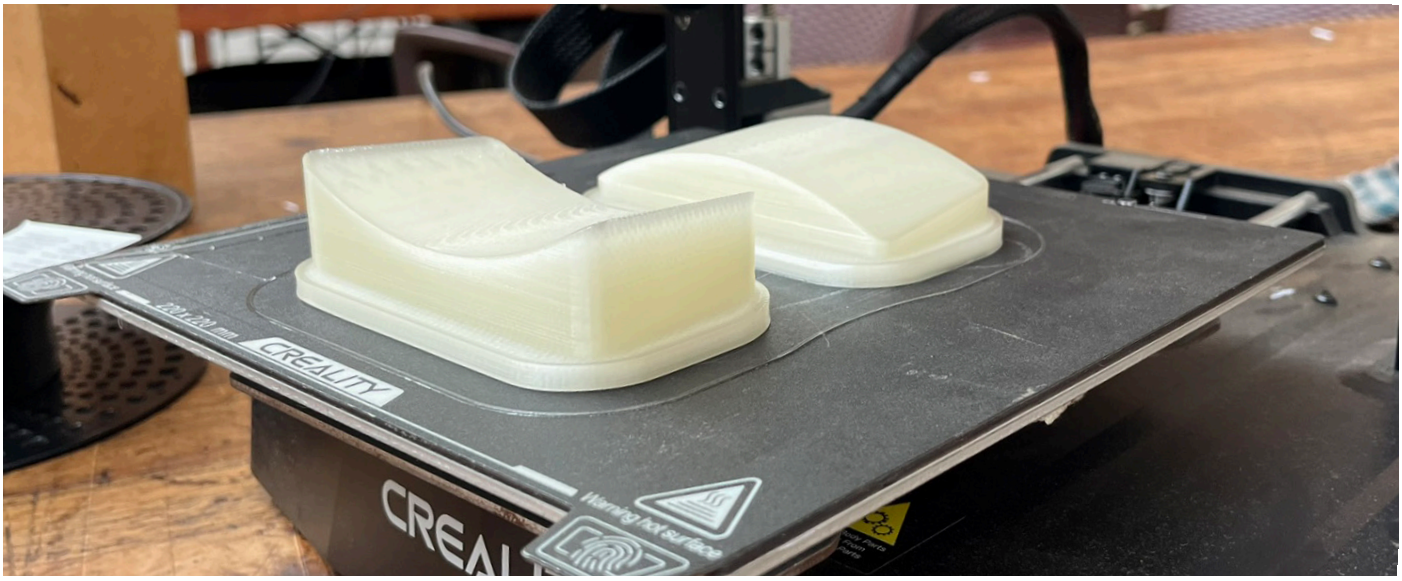
Male mould for seat-plate



Female mould for seat-plate

Step 3

3D printing the moulds



Ready mould for the backrest

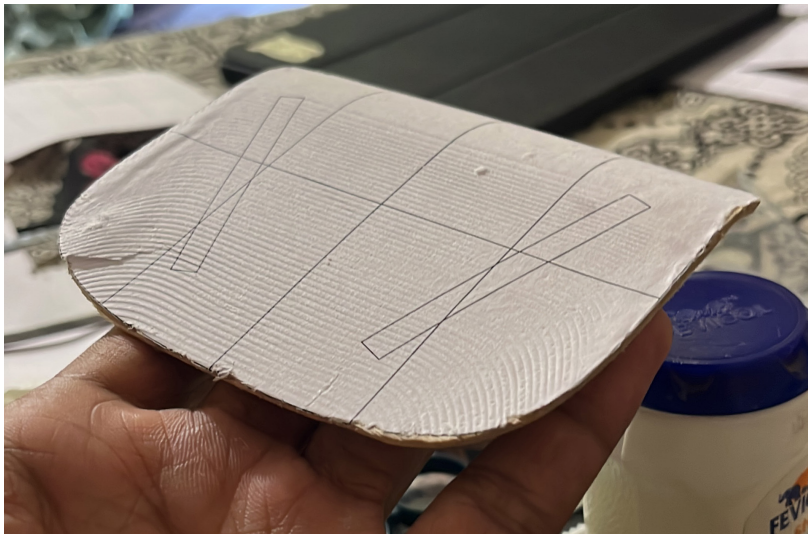
Moulding Components

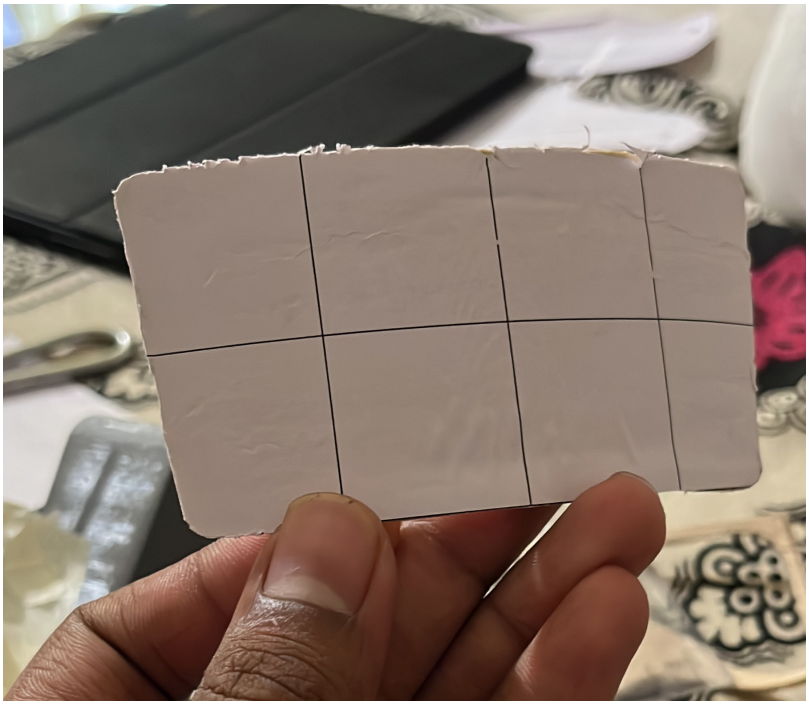
Attempt 1















STEPS

1. Cutting 1mm balsa sheet
2. Soaking in water for 3-4 hrs
3. Applying fevicol sh, and layering sheets
4. Clamping until it dries

PROBLEMS

1. Takes too long to dry
2. When kept open to dry, it deforms

Moulding Components

Attempt 2













STEPS

1. Cutting wooden plank on table saw
2. Sanding the sheets down to 1 mm to create mouldable veneer
3. Using pressure cooker to steam the sheets
4. Layering sheets with Araldite (so it dries faster, sets quicker)

PROBLEMS

1. Takes too long to dry
2. The handmade veneer sheets warp too much while drying

Creating Base







STEPS

1. Creating stencils using the 3d model
2. Applying stencils on wood
3. Cutting according to the stencils using junior saw
4. Sticking extra corners where needed
5. Sanding the fillets

Applying Finish

Attempt 1









STEPS

1. Sanding with 300, 400, 600 grit sandpaper
2. Removing any residue of fevicol remaining after removing the stencil
3. Applying water-based pigment to achieve yellowish color of the original surface
4. Applying PU finish

PROBLEMS

1. Sanding balsa wood, and then applying water based pigment made the grains puff up
2. The PU finish didn't look as intended. Uneven Surface, too glossy

Final Model





Assembled chair without finish









STEPS

1. Remaking the molded components using dry balsa sheet (without soaking)
2. Removing the PU finish from the base frame
3. Assembling and gluing the components together using fevicol sh
4. Applying oil based matte finish



Daystak Side Chair RD1

Furniture Type
Side Chair

Furniture Name
Daystak Side Chair RD1

Original Name
Hillestack Side Chair

Year
1951

Designer
Robin Day

Place of Origin
Hille's Watford facility, UK

Re-issue By
& Tradition (2024)

Material
Beech Wood

Weight
4.5kg

Robin Day



About

Robin Day (1915–2010) was a defining figure of 20th-century British design, celebrated for his elegant, democratic approach to furniture and industrial design. Known especially for the pioneering Polypropylene stacking chair—first released in 1963 and produced in the tens of millions—Day aimed to make good design affordable and functional for everyday environments in schools, airports, offices, and homes

Key Accomplishments

- Winning the MoMA Low-Cost Furniture Competition (with Clive Latimer) in 1948, which launched his collaboration with Hille and a wave of modernist seating and storage designs

- Designing furniture for the Festival of Britain (1951) and the seating for the Royal Festival Hall, widely regarded as emblematic of post-war optimism and functionality

- Championing a design philosophy rooted in simplicity, economy of materials, and long-term social value over fleeting novelty.

& Tradition

&Tradition – founded in Copenhagen in 2010 – is a global design house rooted in Danish craftsmanship. We re-issue iconic classics and develop tomorrow's icons, empowering a network of creatives to shape our story. Guided by originality, sustainability and people-first values, we blend heritage, contemporary production and conscientious practice in every piece.

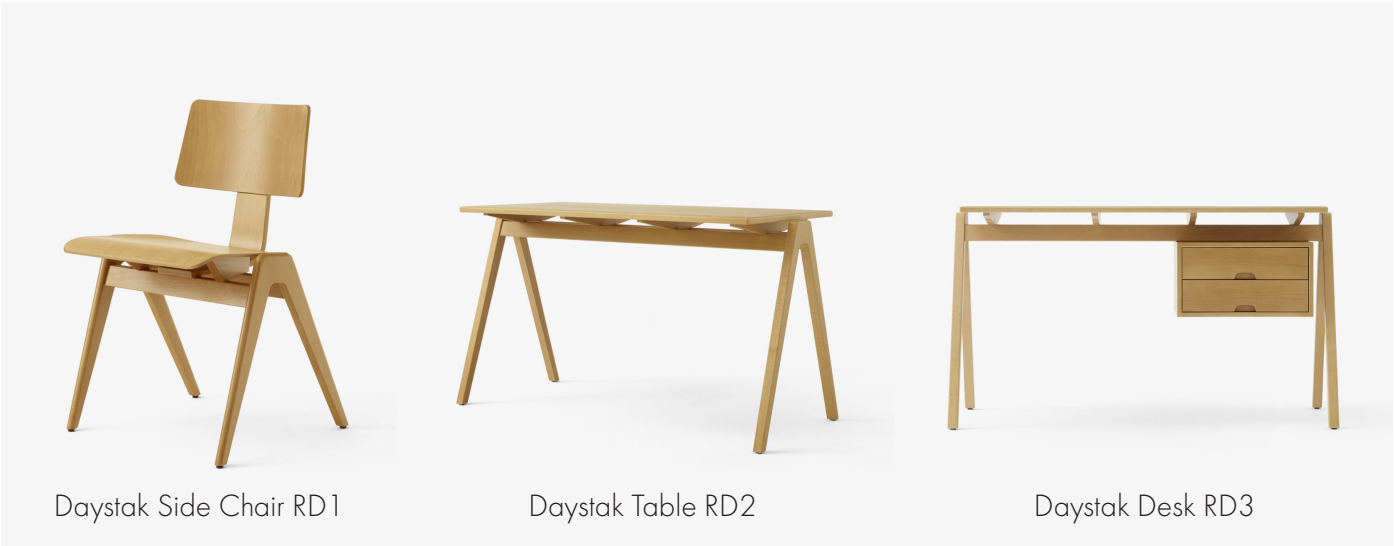
Daystak Range

Original Concept (1951):

Robin Day's first low-cost furniture collection for Hille was originally named Hillestak and included a side chair, table, and desk—all sharing the signature A-shaped beech leg structure and stackable design motif

Rebranded & Relaunched (2024)

Under the partnership between &Tradition and the Robin and Lucienne Day Foundation, these pieces were reintroduced as the Daystak collection, preserving Day's original forms with only minor updates for modern functional standards



Daystak Side Chair RD1

Daystak Table RD2

Daystak Desk RD3

Daystak Side Chair

Context

The utilitarian Hillestak Chair (1951), a stacking chair with a beech plywood seat and a solid beech frame, was his first mass-produced design made for Hille & Co.

The partnership proved so successful that, for the next 20 years, he created most of Hille's designs.

Design Philosophy

Robin Day believed that good design is a social force: it must solve real problems, be rigorously and efficiently constructed and remain affordable for the many. Rooted in post-war austerity, his work champions clarity of purpose and honest expression of construction through mass-produced, low-cost seating that enhances everyday environments.

The chair reflects post-war modernism, availability of new production techniques and materials

Ergonomics

shaped seat and back support made from laminated veneer that flexes subtly for comfort. Carefully proportioned seat height (~46.3 cm) and depth (~37.5 cm) suited for prolonged dining or workspace use.

Salient Features

- Stackable up to 5 chairs high
- Weighs approximately 4.5 kg
- The laminated backrest support member flexes subtly under pressure, providing comfort. Here the material used to the advantage of its physical properties



Front View



Side View



Side Chair Placed With the Daystak RD3 Desk

Daystak Side Chair

Materials Used

Solid beech, beech veneer.

Making Process

- The solid beech frame is cut using a CNC machine.
- The seat and back supports are made by form-pressing laminated veneer layers together with glue, and then cut into their final shape using a CNC machine.
- The entire chair is then sanded and the frame is glued together.
- All the components are assembled using screws.

Materials Used

Solid beech, beech veneer.

Joineries

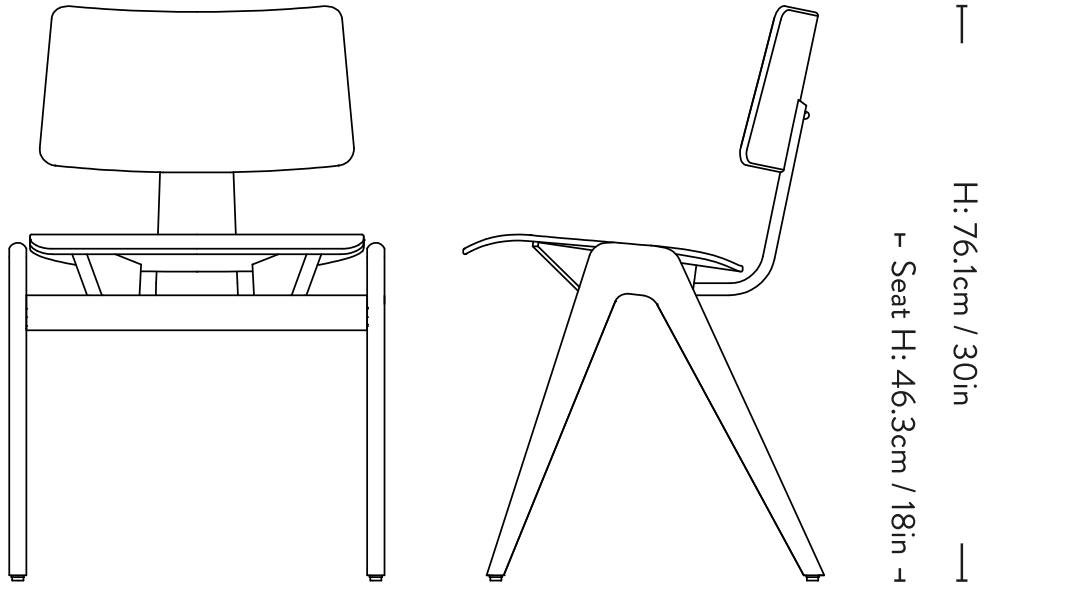
- The frame, seat plate, backrest are assembled together using screws.
- The frame is put together using glue

Finish

- A water-based lacquer, offering a clear, natural-toned protective surface



the surface finish



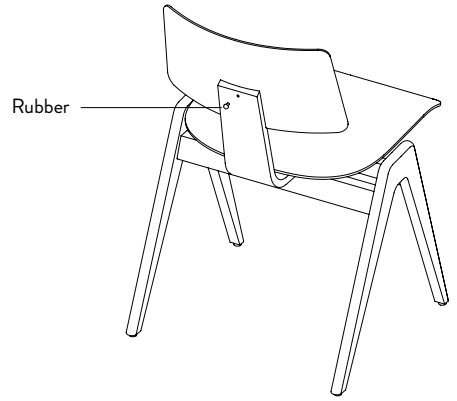
dimensions

Components

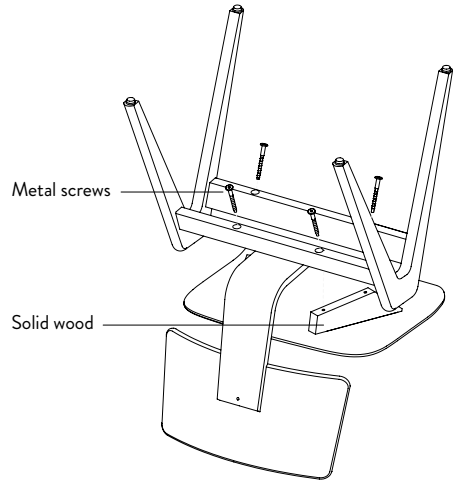
1. Stacking Knob
2. Backrest
3. Seat-Plate
4. Frame
5. Gliders
6. Screws



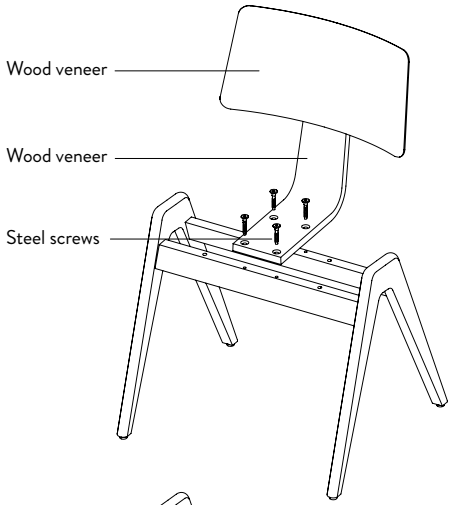
Disassembly Instructions



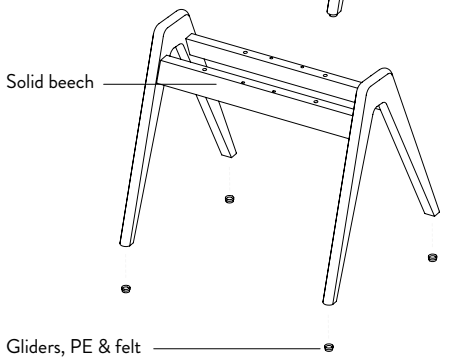
1. Remove the rubber stacking knob using pliers. Do this with caution as it is mounted with glue.



2. Remove the seat plate by unscrewing the four screws beneath the seat.



3. Turn the chair the right way up and unscrew the four screws to seperate the backrest from the base.



4. Remove the gliders using a flat-head screwdrive.

